



**SCHOOL OF
COMPUTING**

DESIGN AND ANALYSIS OF ALGORITHMS
LAB WORKBOOK
WEEK - 7

NAME : SANTHOSH A

ROLL NUMBER : CH.SC.U4CSE24142

CLASS : CSE-B

Question 1: To implement the Greedy algorithm for the Job Sequencing with Deadlines problem and determine the optimal sequence of jobs that maximizes total profit.

Consider a set of 14 jobs, where each job requires one unit of time for completion. Each job has an associated profit and deadline. A job must be completed on or before its deadline to earn the profit.

The profits of the jobs are:

{22, 19, 29, 28, 30, 21, 27, 25, 24, 26, 14, 27, 19, 11}

The deadlines of the jobs are:

{3, 3, 8, 6, 7, 5, 10, 4, 6, 12, 13, 2, 14, 1}

CODE:

```
//CH.SC.U4CSE24142 - SANTHOSH A
#include <stdio.h>
#define MAX 50
struct Job {
    int id;
    int profit;
    int deadline;
};
int main() {
    printf("CH.SC.U4CSE24142 - SANTHOSH A\n");
    int n, i, j;
    printf("Enter number of jobs: ");
    scanf("%d", &n);
    struct Job jobs[MAX];
    for(i = 0; i < n; i++) {
        jobs[i].id = i + 1;
        printf("\nEnter profit for Job %d: ", i+1);
        scanf("%d", &jobs[i].profit);
        printf("Enter deadline for Job %d: ", i+1);
        scanf("%d", &jobs[i].deadline);
    }
    for(i = 0; i < n-1; i++) {
```

```

        for(j = 0; j < n-i-1; j++) {
            if(jobs[j].profit < jobs[j+1].profit) {
                struct Job temp = jobs[j];
                jobs[j] = jobs[j+1];
                jobs[j+1] = temp;
            }
        }
    }

    int maxDeadline = 0;
    for(i = 0; i < n; i++) {
        if(jobs[i].deadline > maxDeadline)
            maxDeadline = jobs[i].deadline;
    }

    int slot[MAX];
    int totalProfit = 0;
    for(i = 0; i <= maxDeadline; i++)
        slot[i] = -1;
    for(i = 0; i < n; i++) {
        for(j = jobs[i].deadline; j > 0; j--) {
            if(slot[j] == -1) {
                slot[j] = jobs[i].id;
                totalProfit += jobs[i].profit;
                break;
            }
        }
    }

    printf("\n\nTime Slot\tJob\n");
    for(i = 1; i <= maxDeadline; i++) {
        if(slot[i] != -1)
            printf("%d\t\tJ%d\n", i, slot[i]);
        else
            printf("%d\t\t--\n", i);
    }
}

```

```

printf("\nMaximum Profit = %d\n", totalProfit);

return 0;
}

```

OUTPUT:

```

D:\AVV CHENNAI\Semester 4\Design and Analysis of Algorithms\Lab Activities\Week 7>gcc Job_Sequencing.c
D:\AVV CHENNAI\Semester 4\Design and Analysis of Algorithms\Lab Activities\Week 7>a
CH.SC.U4CSE24142 - SANTHOSH A
Enter number of jobs: 14

Enter profit for Job 1: 22
Enter deadline for Job 1: 3

Enter profit for Job 2: 19
Enter deadline for Job 2: 3

Enter profit for Job 3: 29
Enter deadline for Job 3: 8

Enter profit for Job 4: 28
Enter deadline for Job 4: 6

Enter profit for Job 5: 30
Enter deadline for Job 5: 7

Enter profit for Job 6: 21
Enter deadline for Job 6: 5

Enter profit for Job 7: 27
Enter deadline for Job 7: 10

Enter profit for Job 8: 25
Enter deadline for Job 8: 4

Enter profit for Job 9: 24
Enter deadline for Job 9: 6

Enter profit for Job 10: 26
Enter deadline for Job 10: 12

Enter profit for Job 11: 14
Enter deadline for Job 11: 13

Enter profit for Job 12: 27
Enter deadline for Job 12: 2

Enter profit for Job 13: 19
Enter deadline for Job 13: 14

Enter profit for Job 14: 11
Enter deadline for Job 14: 1

```

Time Slot	Job
1	J6
2	J12
3	J1
4	J8
5	J9
6	J4
7	J5
8	J3
9	--
10	J7
11	--
12	J10
13	J11
14	J13

Maximum Profit = 292

WORKING:

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LAB WEEK - 7 JOB SEQUENCING

To implement the greedy algorithm for the job sequencing with deadlines and determine the optimal sequence of jobs that maximises the total profit.

Consider a set of 14 jobs, where each job requires one unit of time for completion. Each job has an associated profit and deadline.

The profits of the jobs are:

{ 22, 19, 29, 28, 30, 21, 27, 25, 24, 26, 14, 27, 19, 11 }

The deadlines of the jobs are:

{ 3, 3, 8, 6, 7, 5, 10, 4, 6, 12, 13, 2, 14, 1 }

STEP-1: Arrange the jobs in descending order of profit.

JOB	PROFIT	DEADLINE
J ₅	30	7
J ₃	29	8
J ₄	28	6
J ₇	27	10
J ₁₂	27	2
J ₁₀	26	12
J ₂	25	4
J ₉	24	6
J ₁	22	3
J ₆	21	5
J ₂	19	3
J ₁₃	19	14
J ₁₁	14	18
J ₁₄	11	1

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Since the maximum value of deadline is 14, fourteen time slots are created.

STEP-2: Assign jobs at the optimal time slots.

01) Add $J_5 (30, 7)$

→ J_5 can be added in slot 7 since it is empty.

						J_5									
--	--	--	--	--	--	-------	--	--	--	--	--	--	--	--	--

→ Sequence is $\{J_5\}$ and the profit is 30.

02) Add $J_3 (29, 8)$

→ J_3 can be added in slot 8 since it is empty.

						J_5	J_3								
--	--	--	--	--	--	-------	-------	--	--	--	--	--	--	--	--

→ Sequence is $\{J_5, J_3\}$ and the profit is 59

03) Add $J_4 (28, 6)$

→ J_4 can be added in slot 6 since it is empty.

					J_4	J_5	J_3								
--	--	--	--	--	-------	-------	-------	--	--	--	--	--	--	--	--

→ Sequence is $\{J_5, J_3, J_4\}$ and the profit is 87

04) Add $J_7 (27, 10)$

→ J_7 can be added in slot 10 since it is empty.

					J_4	J_5	J_3		J_7						
--	--	--	--	--	-------	-------	-------	--	-------	--	--	--	--	--	--

→ Sequence is $\{J_5, J_3, J_4, J_7\}$ and the profit is 114

05) Add $J_{12} (27, 2)$

→ J_{12} can be added in slot 2 since it is empty.

J_{12}					J_4	J_5	J_3		J_7						
----------	--	--	--	--	-------	-------	-------	--	-------	--	--	--	--	--	--

→ Sequence is $\{J_5, J_3, J_4, J_7, J_{12}\}$ and the profit is 141

06) Add $J_{10} (26, 4)$

→ J_{10} can be added in slot 4 since it is empty.

J_{12}			J_{10}		J_4	J_5	J_3		J_7		J_{10}				
----------	--	--	----------	--	-------	-------	-------	--	-------	--	----------	--	--	--	--

→ Sequence is $\{J_5, J_3, J_4, J_7, J_{12}, J_{10}\}$ and the profit is 167

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07) Add $J_8(25, 4)$ → J_8 can be added in slot 4 since it is empty.

	J_{12}		J_8		J_4	J_5	J_3		J_7		J_{10}		
--	----------	--	-------	--	-------	-------	-------	--	-------	--	----------	--	--

→ Sequence is $\{J_5, J_8, J_4, J_7, J_{12}, J_{10}, J_3\}$ and the profit is 19208) Add $J_9(24, 6)$ → J_9 cannot be added in slot 6 since it is occupied.

→ But it can be added to slot-5 since it is empty.

	J_{12}		J_8	J_9	J_4	J_5	J_3		J_7		J_{10}		
--	----------	--	-------	-------	-------	-------	-------	--	-------	--	----------	--	--

→ Sequence is $\{J_5, J_8, J_4, J_7, J_{12}, J_{10}, J_8, J_9\}$ and the profit is 21609) Add $J_1(22, 3)$ → J_1 can be added in slot 3 since it is empty.

	J_{12}	J_1	J_8	J_9	J_4	J_5	J_3		J_7		J_{10}		
--	----------	-------	-------	-------	-------	-------	-------	--	-------	--	----------	--	--

→ Sequence is $\{J_5, J_8, J_4, J_7, J_{12}, J_{10}, J_8, J_9, J_1\}$ and the profit is 23610) Add $J_6(21, 5)$ → J_6 cannot be added in slot-5.→ J_6 cannot be added in slot-4.→ J_6 cannot be added in slot-3.→ J_6 cannot be added in slot-2.→ J_6 can be added to slot-1 since it is free.

J_6	J_{12}	J_1	J_8	J_9	J_4	J_5	J_3		J_7		J_{10}		
-------	----------	-------	-------	-------	-------	-------	-------	--	-------	--	----------	--	--

→ Sequence is $\{J_5, J_8, J_4, J_7, J_{12}, J_{10}, J_8, J_9, J_1, J_6\}$ and the profit is 25911) Add $J_2(19, 3)$ → J_2 cannot be added in slot-3.→ J_2 cannot be added in slot-2.→ J_2 cannot be added in slot-1.∴ J_2 is being skipped.

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12) Add $J_{13}(19, 14)$ → J_{13} can be added in slot 14 since it is empty.

J_6	J_{12}	J_1	J_2	J_9	J_4	J_5	J_3		J_7		J_{10}	J_{13}
-------	----------	-------	-------	-------	-------	-------	-------	--	-------	--	----------	----------

→ Sequence is $\{J_5, J_3, J_4, J_7, J_{12}, J_{10}, J_2, J_9, J_1, J_6, J_{13}\}$ and the profit is 278.13) Add $J_{11}(14, 13)$ → J_{11} can be added in slot 13 since it is empty.

J_6	J_{12}	J_1	J_2	J_9	J_4	J_5	J_3		J_7		J_{10}	J_{11}	J_{13}
-------	----------	-------	-------	-------	-------	-------	-------	--	-------	--	----------	----------	----------

→ Sequence is $\{J_5, J_3, J_4, J_7, J_{12}, J_{10}, J_2, J_9, J_1, J_6, J_{13}, J_{11}\}$ and the profit is 292.14) Try adding $J_{14}(11, 1)$ → J_{14} cannot be added since slot-1 is occupied.

FINAL JOB SEQUENCE :-

J_6	J_{12}	J_1	J_2	J_9	J_4	J_5	J_3		J_7		J_{10}	J_{11}	J_{13}
-------	----------	-------	-------	-------	-------	-------	-------	--	-------	--	----------	----------	----------

Using the greedy job sequencing algorithm, the optimal sequence of jobs yields a maximum profit of 292.

Time Complexity: $O(N^2)$: The time complexity of the program is $O(N^2)$ because it uses bubble sort to arrange jobs by profit and a nested loop to assign jobs to time slots in the worst case.

Space Complexity: $O(N)$: The space complexity is $O(N)$ since we use arrays to store the jobs and time slots proportional to the number of jobs.